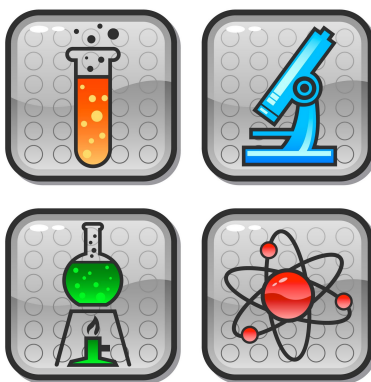




UNIVERSITY INTERSCHOLASTIC LEAGUE

Science

Invitational B • 2021



GENERAL DIRECTIONS:

- **DO NOT OPEN EXAM UNTIL TOLD TO DO SO.**
- Contestants may take up to two hours to complete the contest. If you are in the process of actually writing an answer when the signal to stop is given, you may finish writing that answer.
- Papers may not be turned in until 30 minutes have elapsed. If you finish the test in less than 30 minutes, remain at your seat and retain your paper until told to do otherwise. You may use this time to check your answers.
- All answers must be written on the answer sheet provided. Indicate your answers in the appropriate blanks provided on the answer sheet. Write clearly and legibly!
- You may place as many notations as you desire anywhere on the test paper but not on the answer sheet, which is reserved for answers only.
- You may use additional scratch paper provided by the contest director.
- All questions have ONE and only ONE correct (BEST) answer. There is a penalty for all incorrect answers.
- If a question is omitted, no points are given or subtracted.
- The back two pages of this test include a copy of the periodic table of the elements, as well as listings of other scientific relationships. You may use this information during the contest and may detach the back page from the test if you wish.
- A simple scientific calculator is sufficient for the high school Science contest. **The UIL provides a list of approved calculators that meet the criteria for use in the Science contest. No other calculators are permitted during the contest.** The Science Contest Approved Calculator List is available in the current Science Contest Handbook and on the UIL website. Contest directors will perform a brief visual inspection to confirm that all contestants are using only approved calculators. Each contestant may use up to two approved calculators during the contest.

- B01. Which of the following best describes denitrification?
- A) The reduction of nitrate to nitrite and then to dinitrogen gas.
 - B) The production of ammonium.
 - C) The incorporation of dinitrogen gas to produce ammonium.
 - D) The oxidation of ammonium to nitrate.
 - E) The production of ammonium by decomposers.
- B02. All major animal phyla started appearing in the fossil record about 541 million years ago. The event that accounted for all this animal diversity is known as the
- A) dinosaur extinction event.
 - B) great oxidation event.
 - C) K-T extinction event.
 - D) Cambrian explosion.
 - E) Holocene extinction event.
- B03. A helix-turn-helix motif in a protein is indicative of
- A) a DNA-binding protein.
 - B) a transport protein.
 - C) an enzyme.
 - D) a secreted protein.
 - E) a storage protein.
- B04. The independent evolution of analogous structures in different species is called
- A) divergent evolution.
 - B) adaptive radiation.
 - C) convergent evolution.
 - D) homology.
 - E) atavism.
- B05. Which of the following events is common to fermentation and respiration?
- A) citric acid cycle
 - B) glycolysis
 - C) electron transport chain
 - D) oxidative phosphorylation
 - E) chemiosmosis
- B06. In November 2020, the Centers for Disease Control and Prevention, along with public health officials, began investigating three separate incidences of *Escherichia coli* O157:H7 linked to
- A) deli meats.
 - B) onions.
 - C) spinach.
 - D) ground beef.
 - E) unknown sources.
- B07. A biome that is dominated by warm, wet summers, cool winters, and trees that lose their leaves each year is called a
- A) tropical rainforest.
 - B) temperate deciduous forest.
 - C) taiga.
 - D) tundra.
 - E) desert.
- B08. The most abundant leukocyte in human blood is the
- A) neutrophil.
 - B) basophil.
 - C) eosinophil.
 - D) lymphocyte.
 - E) monocyte.
- B09. A protein that is destined to be modified and exported from the cell would travel a specific pathway. All of the answers below are part of that pathway. Relative to the other answers, the third item that the exported protein would travel through is the/a
- A) *cis* Golgi body.
 - B) transport vesicle.
 - C) rough endoplasmic reticulum.
 - D) *trans* Golgi body.
 - E) secretory vesicle.

- B10. The 50S ribosomal subunit of prokaryotes includes
 A) 46 proteins plus 28S, 5.8S, and 5S rRNA.
 B) 33 proteins plus 18S rRNA.
 C) 33 proteins plus 23S and 5S rRNA.
 D) 20 proteins plus 16S rRNA.
 E) the 30S ribosomal subunit.
- B11. Reproductive isolating mechanisms that occur after fertilization include all of the following except
 A) post-zygotic mechanisms.
 B) hybrid sterility.
 C) nonviability of hybrids.
 D) hybrid breakdown.
 E) temporal isolation.
- B12. All of the following are part of the tRNA structure except
 A) variable loop.
 B) T-loop.
 C) D-loop.
 D) codon loop.
 E) acceptor stem.
- B13. What is the *In* value for a hypothetical diploid somatic cell with 40 total chromosomes?
 A) 10
 B) 20
 C) 30
 D) 40
 E) 80
- B14. If red (R) is dominant to white (r) and blue (B) is dominant to purple (b), what percentage of the offspring from the following genetic cross will be both white and purple?

 BbRr x bbrr

 A) 0%
 B) 25%
 C) 50%
 D) 75%
 E) 100%
- B15. Eudicots, monocots, and magnoliids belong to
 A) Kingdom Plantae.
 B) Anthophyta
 C) Domain Eukarya.
 D) angiosperms.
 E) all of the above.
- B16. Muscle cells contain large calcium storage organelles specifically called
 A) vacuoles.
 B) Golgi bodies.
 C) sarcoplasmic reticula.
 D) rough endoplasmic reticula.
 E) smooth endoplasmic reticula.
- B17. Polymerase is to _____ as peptidyltransferase is to _____.
 A) hydrogen; phosphodiester
 B) peptide; hydrogen
 C) phosphodiester; peptide
 D) peptide; phosphodiester
 E) hydrogen; peptide
- B18. Arginine is an amino acid that bacteria can either biosynthesize using the enzymes encoded by the *arg* operon or they can acquire it from their environment. When arginine is plentiful in the environment, what can be hypothesized about the arginine operon?
 A) The operon is activated.
 B) The operon is derepressed.
 C) The operon is induced.
 D) The operon is repressed.
 E) The arginine biosynthesis enzymes are being produced.

- B19. Arthropods that can transmit pathogens are called
- A) fomites.
 - B) helminths.
 - C) worms.
 - D) vectors.
 - E) protists.
- B20. All of the following use an energy source during transport except
- A) active transport.
 - B) group translocation.
 - C) secondary active transport.
 - D) moving substances against their gradients.
 - E) osmosis.

- C01. Modern pennies are made of 97.5% zinc and 2.5% copper by weight. If a modern penny has a mass of 2.50 grams, how many copper atoms are in a 2021 penny?



- A) 5.9×10^{20} Cu atoms
 B) 3.3×10^{21} Cu atoms
 C) 2.4×10^{22} Cu atoms
 D) 9.7×10^{23} Cu atoms
 E) 1.5×10^{24} Cu atoms
- C02. What is the maximum number of electrons that can be found in the fourth energy level of an atom?
- A) 18
 B) 32
 C) 36
 D) 42
 E) 54

- C03. In any electrochemical cell, which process occurs at the anode?

A) reduction
 B) precipitation
 C) hydration
 D) oxidation
 E) deduction

- C04. When carbon disulfide, CS_2 , is completely combusted, the products are CO_2 and SO_2 . If 2.0 moles of CS_2 are combusted, what is the total mass of the products formed?

A) 344.30 g
 B) 152.30 g
 C) 76.15 g
 D) 248.30 g
 E) 172.15 g

- C05. For the endothermic gas phase equilibrium reaction $\text{A}_2 + 3 \text{B}_2 \rightleftharpoons 2 \text{AB}_3$, which of these actions will result in more product being formed at equilibrium?

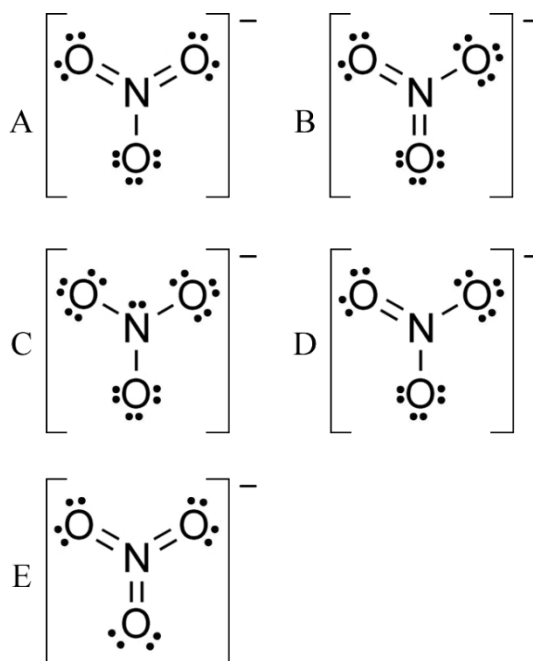
I. Heat the reaction
 II. Shake the container
 III. Reduce the volume of the container

A) I only
 B) I and II
 C) I and III
 D) II and III
 E) III only

- C06. What volume of 0.325 M sulfuric acid is required to neutralize 1.645 g of solid NaOH?

A) 127 mL
 B) 31.6 mL
 C) 198 mL
 D) 63.3 mL
 E) 99.0 mL

- C07. Which of these is the best Lewis dot structure for the nitrate ion?



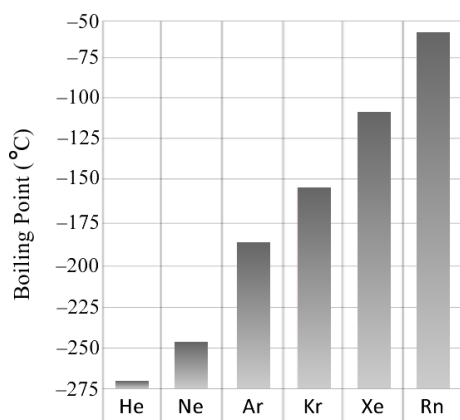
C08. Which of these is *not* an oxidation-reduction process?

- A) A combustion reaction
- B) A precipitation reaction
- C) A decomposition reaction
- D) A synthesis reaction
- E) A single replacement reaction

C09. An atom of a monatomic gas at STP has an atomic mass of 39.962383 Da. How many neutrons are in the nucleus of this atom?

- A) 40
- B) 20
- C) 18
- D) 0
- E) 22

C10. This chart shows the boiling points of the noble gases. How do you explain this trend?



- A) The more electrons an atom has, the more bonds it can form.
- B) Larger atoms with more electrons are more easily polarizable.
- C) The increasing radius leads to greater ionic character in bonds between atoms.
- D) Heavier atoms have a stronger gravitational attraction to earth, making them boil at higher temperatures.
- E) Heavy atoms are inherently warmer internally than light atoms are, so it takes less added heat to make them boil.

C11. What is the change in volume when 2.5 moles of neon gas at 1 atm pressure and 85 °C is cooled to -40 °C and 0.60 atm pressure?

- A) +39 L
- B) -4.8 L
- C) -11 L
- D) +0.82 L
- E) +6.2 L

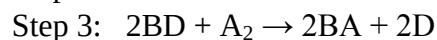
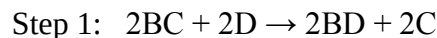
C12. If a physical transformation is endothermic and involves an increase in entropy, when would this transformation be spontaneous?

- A) Always
- B) Never
- C) At higher temperatures
- D) At lower temperatures
- E) Cannot be determined without knowing the equilibrium constant.

C13. The K_{sp} of cadmium arsenate, $Cd_3(AsO_4)_2$, is 2.2×10^{-33} . What is the concentration of Cd^{2+} in a saturated solution of cadmium arsenate?

- A) 3.5×10^{-7} M
- B) 1.2×10^{-7} M
- C) 2.3×10^{-7} M
- D) 2.9×10^{-7} M
- E) 8.8×10^{-7} M

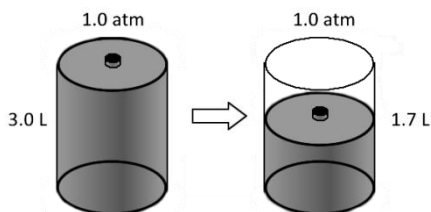
C14. The overall reaction $A_2 + 2BC \rightarrow 2BA + C_2$ proceeds by a series of steps, shown below:



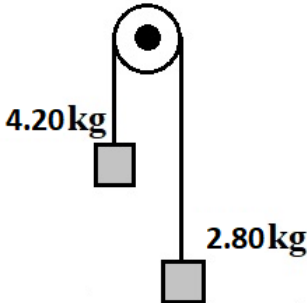
Which species in this reaction mechanism is a catalyst?

- A) A_2
- B) BC
- C) C
- D) D
- E) BD

- C15. How much work is done on the system when a sample of gas is reduced from 3.0 L to 1.7 L against a constant pressure of 1 atm?
1 liter atmosphere = 101.325 joules

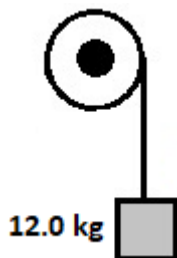


- A) 132 J
B) -132 J
C) 172 J
D) -172 J
E) 152 J
- C16. How many grams of NaCl can you make if you have 10.0 grams of sodium metal and 5.00 L of Cl_2 gas at STP?
- A) 13.0 g
B) 4.12 g
C) 26.1 g
D) 13.8 g
E) 25.4 g
- C17. A student adds excess KCl to a 100 mL solution of silver nitrate and precipitates out 7.6 grams of solid AgCl. What was the concentration of silver nitrate in the solution?
- A) 0.70 M
B) 0.53 M
C) 0.45 M
D) 0.33 M
E) 0.26 M
- C18. Which of these elements in the ground state is an exception to the aufbau principle?
- A) Tin
B) Lead
C) Copper
D) Magnesium
E) Radium
- C19. You have four bottles in front of you, labeled 0.10 M HBrO , 0.10 M HClO , 0.10 M HIO , and $0.10\text{ M HIP}_2\text{O}$. Calculate the pH of the solution that has the lowest pH.
- | | | |
|-------------------|------------------------|-----------------------------|
| hypobromous acid | HBrO | $K_a = 2.0 \times 10^{-9}$ |
| hypochlorous acid | HClO | $K_a = 3.5 \times 10^{-8}$ |
| hypoiodous acid | HIO | $K_a = 2.0 \times 10^{-11}$ |
| hypopotamous acid | HIP_2O | $K_a = 8.8 \times 10^{-13}$ |
- A) 3.75
B) 4.23
C) 4.85
D) 5.85
E) 6.53
- C20. A scientist needs to make 2.00 liters of $0.500\text{ M Na}_2\text{SO}_4$ solution, but she only has 128.56 grams of sodium sulfate in the lab to work with. How much solution can she make?
- A) 1.38 L
B) 1.46 L
C) 1.66 L
D) 1.81 L
E) 2.15 L

- P01. According to Rovelli, which of the following is **not** one of the predictions of Einstein's General Theory of Relativity?
- Planets orbit the Sun because space curves.
 - Light will bend around the Sun.
 - Time speeds up near a planet.
 - Space itself can ripple and sway like the surface of the sea.
 - Space itself can collapse into a bottomless hole.
- P02. According to Rovelli, Einstein furthered the development of quantum theory by proposing that...
- energy is distributed in lumps, or quanta.
 - light is made of packets, called photons.
 - electrons in atoms can only exist at certain energies.
 - objects have no definite position, except during an interaction.
 - we can only calculate the probability of an electron popping up here or there.
- P03. According to Rovelli, in the Greek world view formulated by philosophers such as Pythagoras and Aristotle, the Earth was what geometric shape?
- A flat disk
 - A flat equilateral triangle
 - An irregular stone
 - A cube
 - A sphere
- P04. Ionized hydrogen regions in interstellar space (called H-II regions) are indicative of active star formation. Heated by nearby hot stars, the hydrogen in these regions generally glows what visible color?
- Red
 - Yellow
 - Green
 - Blue
 - White
- P05. For the following calculation, select the answer that has the correct number of significant figures.
- $$\frac{1.98 \times 10^3 - 2.002 \times 10^3}{19.3}$$
- 1
 - 1.1
 - 1.14
 - 1.140
 - 1.1399
- P06. An evil squirrel throws a pecan directly downward with an initial velocity of 5.25m/s. The pecan is thrown from a point that is 2.29m directly above your head. How fast is the pecan moving when it hits the top of your head?
- 6.30 m/s
 - 7.07 m/s
 - 8.51 m/s
 - 9.41 m/s
 - 13.8 m/s
- P07. An Atwood machine is set up with a 4.20kg mass on one side and a 2.80kg mass on the other (as shown). The pulley and rope for this machine are both massless and frictionless. Assuming the masses start from rest, how fast are they moving 1.30seconds after they are released?
- 7.64 m/s
 - 5.88 m/s
 - 5.10 m/s
 - 3.92 m/s
 - 2.55 m/s
- 
- P08. A 300.0g red block slides to the right at a speed of 5.20m/s on a frictionless surface. The red block strikes a stationary green block that has a mass of 420.0g. After the collision, the red block is sliding slowly backward (to the left) at 0.850m/s. How fast (to the right) is the green block sliding after the collision?
- 3.71 m/s
 - 4.32 m/s
 - 6.05 m/s
 - 7.28 m/s
 - 8.47 m/s

- P09. A fixed pulley has a rope wrapped around it and a mass of 12.0 kg attached to the end of the rope (as shown). The pulley has a rotational inertia of $0.220\text{ kg}\cdot\text{m}^2$ and a radius of 30.0 cm . The rope is massless and as it unwinds, it does not slip on the pulley. Once released, what is the linear acceleration of the mass?

- A) 7.58 m/s^2
 B) 8.14 m/s^2
 C) 8.68 m/s^2
 D) 9.24 m/s^2
 E) 9.80 m/s^2



- P10. A strong wind blows across some hollow reeds that stick up from the water at the edge of a nearby lake. As the wind blows, the reeds make a sound, resonating at a frequency of about 280.0 Hz . If we assume that this sound is the fundamental frequency, and if we treat the reeds as open-closed pipes, then what is the height (above the water) of the average reed?

Assume the air temperature is 20°C .

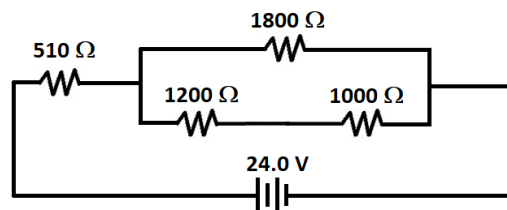
- A) 61 cm
 B) 41 cm
 C) 31 cm
 D) 20 cm
 E) 9.3 cm

- P11. In the summer, when the temperature is 45.0°C , the gap between sections of a bridge is only 0.50 cm . Each bridge section is 12.5 m in length, and the coefficient of thermal expansion of concrete is $1.0 \times 10^{-5}\text{ }^\circ\text{C}^{-1}$. How wide will the gap between the sections be in the winter when the temperature is -10.0°C ?

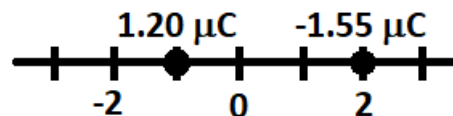
- A) 0.34 cm
 B) 0.59 cm
 C) 0.69 cm
 D) 0.84 cm
 E) 1.2 cm

- P12. For the circuit shown, what is the current flowing through the 1800Ω resistor?

- A) 6.0 mA
 B) 7.2 mA
 C) 8.8 mA
 D) 13 mA
 E) 16 mA

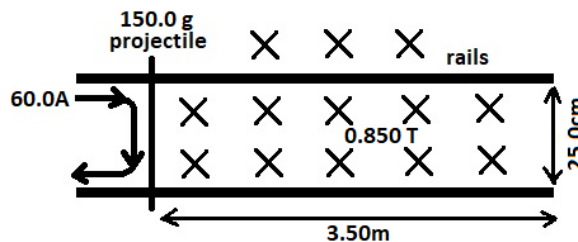


- P13. Two fixed charges are located as shown. A $1.20\mu\text{C}$ charge is at $x = -1.0\text{ cm}$, and a $-1.55\mu\text{C}$ charge is at $x = +2.0\text{ cm}$. What would be the magnitude of the force on a $+0.50\mu\text{C}$ charge placed at the origin ($x = 0$) due to the two fixed charges?



- A) 71.4 N
 B) 56.7 N
 C) 53.9 N
 D) 36.5 N
 E) 17.4 N

- P14. A railgun is constructed as shown, with rails that are 3.50 m long and separated by 25.0 cm . The rails are placed in a magnetic field of 0.850 T . A 150.0 g projectile is set on the rails, and a current of 60.0 A is sent through the projectile. Assuming that it starts from rest, what is the speed of the projectile when it reaches the end of the rails?



- A) 24.4 m/s
 B) 48.8 m/s
 C) 69.0 m/s
 D) 244 m/s
 E) 595 m/s

P15. Unpolarized light travelling in air is incident upon a polished resin tabletop. Using a polarizer, you determine that the light reflected from the tabletop is perfectly polarized when the angle of reflection is 61.0° . What is the index of refraction of the resin tabletop?

- A) 1.18
- B) 1.33
- C) 1.48
- D) 1.61
- E) 1.80

P16. You are using a single converging lens as a magnifier to look at a coin. Unmagnified, the letters on the coin are about 3.00mm long. When you hold the magnifier 4.00cm above the coin, the letters appear to be about 7.00mm long. What is the focal length of the converging lens magnifier?

- A) 2.33 cm
- B) 2.80 cm
- C) 5.60 cm
- D) 7.00 cm
- E) 9.33 cm

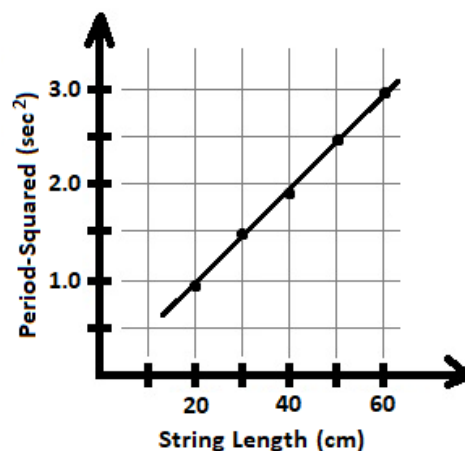
P17. The minimum rest-mass energy uncertainty of a subatomic particle is determined to be 24.0 MeV. Approximately, what is the lifetime of this particle?

- A) about 10^{-26} seconds
- B) about 10^{-23} seconds
- C) about 10^{-20} seconds
- D) about 10^{-18} seconds
- E) about 10^{-12} seconds

P18. While living on Mt. Olympus, you give Zeus a 200.0g sphere of pure Uranium-235 as a birthday gift. Zeus puts the sphere on a bookshelf and forgets about it. After 280 million years, Zeus is cleaning and finds the sphere. How much mass of Uranium-235 remains in the sphere? Note: the half-life of U-235 is 700 million years.

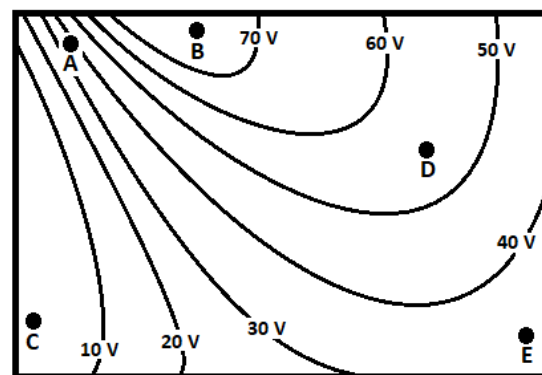
- A) 152g remains
- B) 134g remains
- C) 120g remains
- D) 103g remains
- E) 80.0g remains

P19. While visiting your friend on the planet Solaria, you use a string and a 200.0g mass to make a simple pendulum. You collect data on the period of the pendulum for different lengths of the string. By graphing the period-squared versus the length of the string, you produce the linear plot shown. From this data, determine the acceleration due to gravity on the planet Solaria.



- A) 4.8 m/s^2
- B) 7.0 m/s^2
- C) 8.2 m/s^2
- D) 13 m/s^2
- E) 19 m/s^2

P20. The illustration below represents the equipotential surfaces in a small region of space. At which of the labeled points is the electric field greatest in strength?



- A) point A
- B) point B
- C) point C
- D) point D
- E) point E

Chemistry																1A 1	2A 2	3A 13	4A 14	5A 15	6A 16	7A 17	8A 18
1 H 1.01	2 He 4.00																						
3 Li 6.94	4 Be 9.01																	5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31	3B 3	4B 4	5B 5	6B 6	7B 7	8B 8 9 10			1B 11	2B 12	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95						
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.64	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80						
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.60	53 I 126.90	54 Xe 131.29						
55 Cs 132.91	56 Ba 137.33	57 La 138.9	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.20	83 Bi 208.98	84 Po (209)	85 At (210)	86 Rn (222)						
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (266)	107 Bh (264)	108 Hs (277)	109 Mt (268)	110 Ds (281)	111 Rg (281)	112 Cn (285)	113 Nh (286)	114 Fl (289)	115 Mc (289)	116 Lv (293)	117 Ts (293)	118 Og (294)						

58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)

Water Data

T_{mp}	$= 0^{\circ}\text{C}$
T_{bp}	$= 100^{\circ}\text{C}$
c_{ice}	$= 2.09 \text{ J/g}\cdot\text{K}$
c_{water}	$= 4.184 \text{ J/g}\cdot\text{K}$
c_{steam}	$= 2.03 \text{ J/g}\cdot\text{K}$
ΔH_{fus}	$= 334 \text{ J/g}$
ΔH_{vap}	$= 2260 \text{ J/g}$
K_f	$= 1.86 ^{\circ}\text{C}/m$
K_b	$= 0.512 ^{\circ}\text{C}/m$

Constants

R	$= 0.08206 \text{ L}\cdot\text{atm}/\text{mol}\cdot\text{K}$
R	$= 8.314 \text{ J}/\text{mol}\cdot\text{K}$
R	$= 62.36 \text{ L}\cdot\text{torr}/\text{mol}\cdot\text{K}$
e	$= 1.602 \times 10^{-19} \text{ C}$
N_A	$= 6.022 \times 10^{23} \text{ mol}^{-1}$
k	$= 1.38 \times 10^{-23} \text{ J/K}$
h	$= 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$
c	$= 3.00 \times 10^8 \text{ m/s}$
R_H	$= 2.178 \times 10^{-18} \text{ J}$
m_e	$= 9.11 \times 10^{-31} \text{ kg}$

There is no test-specific data
for this exam.

Physics

Useful Constants

quantity	symbol	value
Free-fall acceleration	g	9.80 m/s^2
Permittivity of Free Space	ϵ_0	$8.854 \times 10^{-12} \text{ C}^2/\text{Nm}^2$
Permeability of Free Space	μ_0	$4\pi \times 10^{-7} \text{ Tm/A}$
Coulomb constant	k	$8.99 \times 10^9 \text{ Nm}^2/\text{C}^2$
Speed of light in a vacuum	c	$3.00 \times 10^8 \text{ m/s}$
Fundamental charge	e	$1.602 \times 10^{-19} \text{ C}$
Planck's constant	h	$6.626 \times 10^{-34} \text{ Js}$
Electron mass	m_e	$9.11 \times 10^{-31} \text{ kg}$
Proton mass	m_p	$1.67265 \times 10^{-27} \text{ kg}$ 1.007276 amu
Neutron mass	m_n	$1.67495 \times 10^{-27} \text{ kg}$ 1.008665 amu
Atomic Mass Unit	amu	$1.66 \times 10^{-27} \text{ kg}$ $931.5 \text{ MeV}/c^2$
Gravitational constant	G	$6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$
Stefan-Boltzmann constant	σ	$5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4$
Universal gas constant	R	$8.314 \text{ J/mol} \cdot \text{K}$ $0.082057 \text{ L} \cdot \text{atm/mol} \cdot \text{K}$
Boltzmann's constant	k_B	$1.38 \times 10^{-23} \text{ J/K}$
Speed of Sound (at 20°C)	v	343 m/s
Avogadro's number	N_A	$6.022 \times 10^{23} \text{ atoms/mol}$
Electron Volts	eV	$1.602 \times 10^{-19} \text{ J/eV}$
Distance Conversion	miles $\rightarrow$ meters	1.00 mile = 1609 meters
Rydberg Constant	R_∞	$1.097 \times 10^7 \text{ m}^{-1}$
Standard Atmospheric Pressure	1 atm	$1.013 \times 10^5 \text{ Pa}$
Density of Pure Water	ρ_{water}	1000.0 kg/m^3